CALLBOX · PILOT BUILD · 13 OCTOBER 2026

Three pilots locked, and CallBox's first P-Series driver

Fast Dial, Micromart and Axe asked for changes before their pilots carry on tomorrow. Every one of them is built, tested and deployed. Axe's CallBox now drives its Yeastar P550 through the P-Series OpenAPI. The phone settings became a **registry of calling drivers and engines**, so a lender switches between S-Series, P-Series and, next, Linkus and Voiso with one choice, and loses nothing when it switches back. Fast Dial's and Micromart's S-Series calling is untouched.

For
The founder, and the next build session

Covers
Fast Dial · Micromart · Axe · Phase 2 Plan B

Code
connected-suite, commit c493810 + 32d830b

Status
Live in production, 13 Oct, 22:28 EAT

1 What each company asked for, and what shipped

COMPANY	ASKED FOR	WHAT CHANGED
Fast Dial, Axe	"Read more" instead of "Open" on the sign-in page, linking to the lender's website.	Each lender's door can now carry its website. Fast Dial's goes to fastdial.co.ke, Axe's to axemicroenterprise.com, in a new tab so the sign-in stays open. The photograph still opens full-size when clicked. Micromart and Mular keep "Open".
Fast Dial	An account dispositioned as a PTP stays on today's list, so follow-ups are hard to manage.	Three causes, all fixed (section 2). A promise now ends the campaign's work on that customer wherever it is saved. The dialler does not hand back anyone who promised today. The agent's home stops pushing them as the next action.
Micromart	Agents see only their own live floor and wallboard.	An agent's live floor shows their own cases, money and calls; a supervisor's shows the whole floor. An agent's wallboard shows their own numbers and seat.
Micromart	No re-assign or escalate in the agent portal.	Both are gone from an agent's case file, and the server refuses them from an agent as well. Re-assign was already supervisor-only on the server; escalate is now too.
Micromart	Callbacks & tasks, promises and recoveries only for the agent's buckets.	Each of the three pages shows an agent their buckets (CollectBox's own `LoanCategories.Agents` teams) plus anything that is theirs by name. The page says which buckets. Supervisors are unchanged.
Micromart	Overview opens on SME, with Fintech as the second choice.	CallBox's Overview now opens on the SME book (3002), with Fintech one click away. The Lending Console keeps its own default.
Axe	Lending console and CallBox in the navy of their website instead of green.	axemicroenterprise.com's own colours, navy `#001247` and orange `#ff8a00`, are now Axe's brand in the LMS. The sign-in page's right side, both console books and the static paint all use them (section 4).

COMPANY	ASKED FOR	WHAT CHANGED
Axe	Connect Axe's CallBox to its P-Series. Calls, auto-dialling, barge, listen, whisper and ending an agent's call should all work. Make the driver a dropdown choice.	A new `yeastar-p` driver, verified against Axe's P550 read-only (section 5). A new supervisor panel lets a supervisor into any call on the PBX, and End now ends the real call. The settings screen offers every driver and engine from one registry (section 6).

2 Fast Dial: why a promised account stayed on the list

Three things kept a customer in front of the agent after they had promised to pay.

1. **The campaign kept them.** Classic CallBox returns a row to the pool only when its outcome is saved *from the campaign*. A promise taken from the work queue, a search, the agent's home or the dialler's promise-day call left the row locked in its campaign, so it was still on the campaign's list. Now a promise completes the campaign's work on that customer wherever it is saved: the row goes back to the pool exactly as it would from the campaign. Other outcomes saved outside a campaign still leave it alone. (lib/callbox/write.ts)
2. **The dialler handed them back.** A promise taken for today counted as a "promise due today", so the dialler offered the same customer again about 45 minutes later. The contact policy now holds back anyone who promised *today*, for campaign, promise-day and broken-promise calls. A callback the agent booked and a customer who rings in still go through. The evidence is a call logged today that carries a promised amount, which both CallBox and classic CallBox write. `promisedtopay.updateddate` was rejected as evidence because classic CallBox stamps it on old promises it closes. (lib/desk/dialler/rules.ts, lib/callbox/contact.ts)
3. **The agent's home pointed back at them.** Its "next best action" picked promises due today, including one taken minutes earlier. Those are skipped now; they still appear in the day's promises.

Checked by `npm run test:dialler`: 54 pass, including seven new cases for "promised today".

3 Micromart: who sees what

SCREEN	SUPERVISOR	AGENT
Live floor	The whole floor, as before	Their own cases by bucket, money credited to them, calls they logged, their own row and activity
Wallboard	The floor	Their numbers and their own seat
Callbacks & tasks	All	Tasks on loans in their buckets, plus the ones they booked
Promise board	All, with the floor's figures	Promises in their buckets plus the ones they took, with figures for that cut
Recoveries	All; may pick one agent	Money recovered out of their buckets, plus money credited to them
Case file	Re-assign, escalate	Neither
Overview	Opens on SME; Fintech is the second chip	

A **bucket** is one of CollectBox's categories (Prepayment, Due, Watch 1–3, NPL, Watch 1 Matured). An agent's buckets are the ones whose team in `LoanCategories.Agents` names them, the same list Micromart's 01:30 run allocates by. An agent on no team sees only what is theirs by name. A login not linked to a CollectBox agent sees nothing, and the page says how to link it (Settings → Seats).

4 Axe's brand

The colours come from axemicroenterprise.com's own stylesheet: navy `#001247` for its header, impact band and "How it works" section, and orange `#ff8a00` for its figures and Contact button. The LMS `Org` row for Axe was changed with `scripts/set-org-brand.ts`; the old green values are kept in the audit line. The two console books now read navy/orange (Boresha) and the site's second navy with its amber (Stawi), so they stay distinguishable. The inside of the desk keeps CallBox's own identity, as every lender's does.

5 Axe's P-Series, as read tonight

Every read below came through the driver CallBox now runs, or the same OpenAPI. Nothing on the PBX was changed, and each test token was given back.

The PBX	Yeastar P550, firmware 37.24.0.73-beta2, at pbx.axemicroenterprise.co.ke:8088. API sign-in in 224–635 ms from Nairobi.
Extensions	15: 1000–1013 and 8000. In the last day agents used 1003–1008, 1012 and 1013. 1000–1002, 1009–1011 and 8000 were idle.
Trunk and routes	One trunk, SAFARICOM-SIP (peer, caller ID +254709478910). The default route takes any number; agents dial `2547…`, which is what CallBox already sends.
Call records	91,388. The ten newest were read back through the driver, and each verdict matched the PBX: ANSWERED 470 s, BUSY, NO ANSWER.
Recordings	**None.** The PBX's recording list is empty, so Axe's P-Series calls are not being recorded. Auto-recording has to be switched on in the PBX (Call Features → Recording) before CallBox can play anything back.
Webhooks	One configured, ConnectBoxAPI's (events 30007, 30008, 30011, 30016). The PBX allows five; CallBox's can be added beside it without touching ConnectBox's.
Reachability	From the internet only TCP 8088 (web and API) answers. SIP on UDP/TCP 5060, 5061, Linkus 8111 and 443 are silent. Click-to-call from the cloud works. MicroSIP can register only from Axe's office network unless remote SIP is opened. The Linkus apps work from anywhere.

Four facts the driver is built around

- **The OpenAPI signs in with the API application, not the admin login.** The username/password are the Client ID and Client Secret under Integrations → API. The admin password cannot be used. The secret first sent was the webhook's signing secret, a different value.
- **Each application may hold at most eight live tokens.** CallBox mints one token per floor, shares it through the database, renews it with its refresh token and gives it back when settings change. It never uses ConnectBox's Linkus SDK key, so it cannot take a token from Axe's live agents.
- **The PBX serves its certificate without the intermediate.** Browsers repair this; Node refuses. The driver fetches the missing GoGetSSL intermediate from the address in the certificate itself and verifies the full chain and the name. Verification is never switched off. Settings → Test stores the intermediate.
- **Calls are controlled by channel, not by extension.** Every P-Series control (hang up, hold, mute, transfer, listen, whisper, barge) names the call channel of one member. The driver finds the channel from the PBX's list of calls in progress when it acts, and follows each call by its own id: from that list while it is up, and from its call record once it ends.

What a P-Series floor can do now

CAPABILITY	HOW
Click-to-call	`call/dial`: the PBX rings every device signed in as the agent's extension (MicroSIP, desk phone, Linkus), then the customer. The auto-dialler places its calls the same way.
Auto-dialler on Axe	New: the dialler draws from the agent's own ConnectBox book: callbacks that have come due, then accounts nobody has worked today, biggest balance first. Only a seat linked to its ConnectBox agent is given work, so nobody can auto-dial Axe's whole book by accident. Axe's agents need CallBox logins and seats for this, which `scripts/import-connectbox-people.ts --commit` creates.
Hang up · hold · mute · transfer	On the agent's own channel. Blind transfer moves the customer's leg with the agent's dial permission.

CAPABILITY	HOW
Listen · whisper · barge	`call/listen` with the agent's channel: the supervisor's own extension rings and joins in that mode.
End an agent's call	New, P-Series only. The command centre's End now ends the real call, and the new "On the PBX right now" panel ends any busy extension's call, after a confirmation.
Supervise any call on the PBX	New panel on Command centre and Phone floor. It shows every busy extension, whether the call came from CallBox, ConnectBox's Linkus, MicroSIP or a desk phone, with Listen, Whisper, Barge and End.
Call state	Dialling (the agent's phone ringing) → ringing (the customer's) → talking → ended with talk time, read from the PBX, not inferred.
Live events (optional)	A receiver at `/api/desk/pbx/yeastar-p/<org>` takes events 30011 and 30012, signed with X-Signature (base64 HMAC-SHA256). It accepts nothing until this floor's own webhook secret is saved.

6 The calling architecture: drivers and engines

A floor's calling is now two separate choices, both drawn from one registry (`lib/desk/telephony/drivers.ts`). The settings screen is built from that registry.

DRIVER: WHAT CALLBOX DRIVES

- **Yeastar S-Series**: live. Fast Dial and Micromart.
- **Yeastar P-Series**: live. Axe.
- **Voiso**: next. For Mular, whose agents sit in Voiso.
- **Direct**, CallBox's own Asterisk 22 voice node: planned (Phase 2, Plan A).
- **Simulated**: live, for training and demos.

ENGINE: HOW THE AGENT HEARS IT

- **The agent's own phone** (MicroSIP, desk handset, Linkus app): live on S and P.
- **Browser through the media edge**: live on S.
- **Browser through Linkus SDK**: next, on P. No edge, works from home.
- **Voiso's agent panel**: next.
- **Browser on the voice node**: planned.

- **Switching is one choice and a Save.** Each driver keeps its own settings in the vault (the P-Series ones under `pseries`, apart from the S-Series fields), so switching driver and back loses nothing. A driver marked next or planned is shown but cannot be switched on until it ships.
- **S-Series floors behave exactly as before.** A configuration saved before engines existed runs the browser phone where a media edge is set, and the agent's own phone otherwise. The S-Series driver file is unchanged.
- **Every driver states its capabilities**, shown as chips under the driver. Screens offer an action only where the driver has it: End on an agent's call and the PBX panel appear only for the P-Series.
- **Settings hardened in passing.** Your Axe screenshot showed the browser filling your sign-in email and password into the TURN fields. Every password field on that screen now refuses autofill. On an S-Series floor the same autofill could have overwritten the PBX API password on Save.

Next: the Linkus engine (P-Series), then Voiso (Mular)

Linkus. ConnectBoxAXE already shows the working sequence on Axe's PBX: `get_token` with the Linkus SDK key, then `sign/create` {username: extension, sign_type: "sdk"}, then `ys-webrtc-sdk-core` `init({username, secret: sign, pbxURL})`. In CallBox the sign is minted server-side for the signed-in agent only. The Linkus SDK key goes in the P-Series settings beside the API one. The engine then becomes selectable on the same dropdown. Its tokens count against that application's own eight, which ConnectBox also uses, so CallBox must share one per floor exactly as the API driver does.

Voiso. The driver slot, engine and capabilities are declared. Building it needs Mular's Voiso API key and a cluster URL: click-to-call, CDR and webhooks behind the same buttons, with supervision staying in Voiso's own panel.

7 To make the first P-Series call

- [ ] On the PBX: choose an idle extension for yourself, for example **1010**, and copy its registration password (Extension → 1010 → edit).
- [ ] MicroSIP: server/domain = the PBX's office address. From outside Axe's office SIP is not reachable, so either sit on Axe's office network or answer on the Linkus app signed in as 1010. Click-to-call rings every device signed in as the extension.
- [ ] **Already done tonight:** Axe's floor is on the P-Series, and your login (Birgen Krosovic) has a supervisor seat on 1010. In CallBox at connectdesk.servicesuitecloud.com/axe, open an account and press Call: your phone rings first, then the customer. Change the extension in Settings → Seats.
- [ ] To let Axe's agents call and auto-dial from CallBox: run `scripts/import-connectbox-people.ts --org axe --commit`. It creates their logins and seats, linked to their ConnectBox ids and extensions. Then switch on auto-dial in Dialler settings.
- [ ] Optional, for live events: on the PBX, Integrations → API → Webhook Event Push → Add the address shown in Settings → Phone system with events 30008, 30011 and 30012, then paste its secret into CallBox.
- [ ] Optional, for playback: switch on recording on the PBX. Today it records nothing.

8 How it was checked

- `npm run test:telephony`: 42 pass. Registry and engine rules, the P-Series' reading of its calls, call records and webhook signatures, number format, and Micromart's bucket scope.
- `npm run test:dialler`: 54 pass, including the seven new "promised today" cases.
- `npm run probe:pseries`: the real driver against Axe's P550, read-only; output in section 5.
- Full TypeScript check and production build: clean. Lint: no new findings.
- Not yet heard end to end: a P-Series click-to-call. It needs a phone signed in as an Axe extension, which only you can set up (section 7).